M13 Batch 3 — Full Set

M13.3 Autoflight (ATA 22) · PDF pages 156–231

Vertical modes, altitude acquire and hold, VNAV and LNAV, localizer and back beam, approach and glideslope, flare, rollout and go-around.

272 questions · Level 3: 102 · Level 2: 121 · Level 1: 49

● marks a question built on a fact your instructor highlighted or underlined in the manual (211 of 272). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

Vertical speed mode

PDF pages 156–160 · 18 questions

LEVEL 1 — RECALL

1. Which quantity does the vertical speed mode bracket and maintain ●
   a. A climb or descend rate
   b. A pitch attitude value
   c. A flight path bank angle

2. Which system safeguards airspeed control automatically while the vertical speed mode is bracketing a rate ●
   a. The autothrust system
   b. The flight management system
   c. The inertial reference system

3. How many control loops make up the cascade used for closed loop control of vertical speed ●
   a. One
   b. Two
   c. Three

4. An autopilot is holding a rate of climb in vertical speed mode. Which unit supplies the controlled value used by the outer loop? ●
   a. The flight management system
   b. The inertial reference system
   c. The air data computer

5. Which kind of controller is the vertical speed controller
   a. A D-controller
   b. A PI-controller
   c. A P-controller

LEVEL 2 — UNDERSTAND AND EXPLAIN

6. What decides how many ways there are of selecting the vertical speed mode on a given aircraft
   a. The transition altitude in force
   b. The vertical speed value stored
   c. The type of autopilot installed

7. Where the vertical speed mode is a basic mode, which condition alone maintains the momentary rate of climb ●
   a. The autopilot is disengaged
   b. The autopilot is engaged
   c. A target flight level is dialled up

8. The autopilot is holding a vertical speed as a basic mode and the crew want a different rate. How is a new reference value introduced? ●
   a. By pressing the vertical speed button on the control panel
   b. By rotating the vertical speed knob on the control panel
   c. By disengaging and re-engaging the autopilot at the new rate

9. How is the vertical speed mode brought in where it is an upper mode rather than a basic one ●
   a. It is maintained automatically by engaging the autopilot alone
   b. It is selected explicitly at the control panel
   c. It is armed by the air data computer at take off

10. When may the reference rate be dialled up in the selection sequence for the vertical speed mode as an upper mode ●
    a. Only before mode activation
    b. Only once the target level is reached
    c. Either before or after mode activation

11. At what point is the outer control loop closed during vertical speed control ●
    a. When the mode is selected
    b. When the capture point is reached
    c. When the target level is reached

12. From where does the outer loop of the rate of climb mode obtain its reference value ●
    a. The flight management computer
    b. The vertical speed memory
    c. The altitude memory

13. With what is the commanded rate of climb compared to give the output error of the outer loop in the rate of climb mode ●
    a. The pitch attitude measured by the inertial reference system
    b. The target flight level set on the autoflight control panel
    c. The actual vertical speed measured by the air data computer

14. What does the vertical speed controller determine and pass on to the inner loop ●
    a. A vertical speed reference value
    b. A pitch attitude reference value
    c. An elevator deflection angle

LEVEL 3 — APPLY AND REASON

15. An upper pitch mode is switched off in flight on an aircraft whose basic pitch mode holds the momentary rate of climb. Which mode does the autopilot then adopt? ●
    a. Altitude acquisition mode
    b. Altitude hold mode
    c. Vertical speed mode

16. The vertical speed mode is an upper mode on the aircraft. Which action comes first in the sequence of events to be observed? ●
    a. Pressing the vertical speed button
    b. Selecting the target flight level
    c. Arming the altitude acquisition mode

17. What is synchronisation of the vertical speed reference value intended to prevent
    a. Controller activity at the moment of mode activation
    b. A reference value being changed after mode activation
    c. A loss of the air data computer rate signal in flight

18. In the vertical speed mode the pitch attitude is altered. Why does the actual rate of climb then change?
    a. Because the trim integrator alters the stabiliser setting
    b. Because the air data computer re-corrects its altitude output
    c. Because the flight path angle changes at constant airspeed

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Answers — Vertical speed mode

1. **a** The mode brackets and holds a vertical speed, which is the same thing as a climb or descend rate. (L1, PDF p. 156)
2. **a** The autothrottle/autothrust system is what keeps the airspeed safe while pitch holds the rate. (L1, PDF p. 156)
3. **b** The cascade for vertical speed has two loops. (L1, PDF p. 160)
4. **c** Vertical speed is an air data parameter and comes from the air data computer. (L1, PDF p. 160)
5. **b** The vertical speed controller is a PI-controller. (L1, PDF p. 160)
6. **c** How many ways there are depends on which autopilot is fitted to the aircraft. (L2, PDF p. 156)
7. **b** As a basic mode it needs no separate selection, so engaging the autopilot alone holds the rate the aircraft already has. (L2, PDF p. 156)
8. **b** With the mode already running, turning the knob is what changes the reference value, and it may be done at any time. (L2, PDF p. 156)
9. **b** An upper mode is not automatic, so the crew must select it deliberately on the panel. (L2, PDF p. 156)
10. **c** The reference value may be set on either side of mode activation. (L2, PDF p. 156)
11. **a** Selecting the mode is what closes the outer loop for vertical speed control. (L2, PDF p. 160)
12. **b** The stored reference for this mode is held in, and supplied from, the vertical speed memory. (L2, PDF p. 160)
13. **c** An error is reference minus actual, and the actual rate is the air data computer measurement. (L2, PDF p. 160)
14. **b** The outer loop controller sets an attitude reference which the inner loop then follows. (L2, PDF p. 160)
15. **c** Where vertical speed is the basic mode, switching off any upper vertical mode leaves the autopilot in vertical speed. (L3, PDF p. 156)
16. **b** The printed sequence begins with selection of the wanted target flight level, before the mode is activated. (L3, PDF p. 156)
17. **a** A stored value unlike the actual rate would make the controller act the instant the mode came in, and synchronisation removes that step. (L3, PDF p. 158)
18. **c** With speed held, a change of attitude becomes a change of flight path angle and therefore of rate of climb or descent. (L3, PDF p. 160)

Level off, mode reversion and altitude acquire

PDF pages 162–168 · 32 questions

LEVEL 1 — RECALL

19. To what value is the vertical speed reference set when the level off mode is selected ●

- **a.** The momentary vertical speed
- **b.** The last value held in memory
- **c.** A rate of zero

20. What does the mode annunciation read while the level off mode is in use ●

- **a.** LVL CHG
- **b.** V/S = 0
- **c.** ALT

21. An altitude has just been pre-selected at the autoflight control panel. What does the autoflight system annunciate? ●

- **a.** ALT active
- **b.** ALT ACQ armed
- **c.** ALT ACQ active

LEVEL 2 — UNDERSTAND AND EXPLAIN

22. The level off mode has just been selected during a climb. What does the vertical speed controller do? ●

- **a.** It reduces the actual rate to zero and holds zero
- **b.** It reduces the actual rate by half and then holds it
- **c.** It holds the altitude reached when the mode was selected

23. What is true of the provisions protecting against violation of airspeed and Mach limits ●

- **a.** They differ a lot between aircraft types
- **b.** They apply only to the vertical speed mode
- **c.** They are identical in every modern autoflight system

24. While a commanded climb rate is being held, over what does the task of the pitch channel take precedence

- **a.** Track controlled by the roll channel
- **b.** Speed controlled by the autothrust system
- **c.** Altitude controlled by the altitude memory

25. Into which altitude does the altitude acquisition mode transfer an aircraft at the end of a climb or descent ●

- **a.** A pilot selected capture altitude or transition level
- **b.** A pilot selected vertical speed or flight path angle
- **c.** A pilot selected QNH-altitude or pressure altitude

26. When is the capture detector of the altitude acquisition mode activated ●

- **a.** When an altitude is pre-selected on the panel
- **b.** When the aircraft reaches the capture point
- **c.** When the altitude hold mode becomes active

27. Together with the actual rate of climb, what else enters the continuous calculation of the capture point ●

- **a.** The barometric correction applied by the air data computer
- **b.** The momentary airspeed error against the speed limit
- **c.** The momentary vertical distance to the selected altitude

28. In which case is the altitude representing the capture point baro-corrected ●

- **a.** When it is at or below the transition altitude
- **b.** When it is anywhere above the transition altitude
- **c.** When it is at or above the selected flight level

29. The aircraft reaches the calculated capture point in a descent. What does the mode logic do?

- **a.** It makes the altitude acquisition mode active
- **b.** It arms the altitude acquisition mode
- **c.** It makes the altitude hold mode active instead

30. At what point is the outer control loop closed in the altitude acquisition mode ●

- **a.** When the capture condition is met
- **b.** When an altitude is pre-selected
- **c.** When the tolerance band is reached

31. To what does the vertical speed reference value correspond in the altitude acquisition outer loop ●

- **a.** The altitude error
- **b.** The measured vertical speed
- **c.** The selected altitude

32. With what is the pre-selected altitude compared in S3 to form the altitude error ●

- **a.** The aircraft's actual altitude
- **b.** The baro-corrected capture altitude
- **c.** The aircraft's actual vertical speed

33. An altitude acquisition is running and the altitude error has become the reference for the outer loop. With what is that reference compared in S2? ●

- **a.** The actual vertical speed
- **b.** The pre-selected altitude
- **c.** The pitch reference value

LEVEL 3 — APPLY AND REASON

34. A pilot selects the level off mode. Why does this mode need no control loop of its own? ●

- **a.** The autothrust system is used with a reference of zero
- **b.** The altitude controller is used with a reference of zero
- **c.** The vertical speed controller is used with a reference of zero

35. An aircraft climbing in vertical speed mode with a very high commanded rate reaches its lower speed limit. What happens to the pitch channel?

- **a.** It reduces the commanded vertical speed to zero and levels off
- **b.** It switches over from vertical speed to airspeed/Mach control
- **c.** It hands pitch control over to the autothrust system

36. The pitch channel has just reverted to speed control during a climb. What does the autothrust system do at the same moment?

- **a.** It remains in speed mode and raises the commanded airspeed
- **b.** It changes from thrust mode to speed mode and holds the commanded airspeed
- **c.** It changes from speed mode to thrust mode and sets a thrust limit

37. After a reversion, to which mode does the combination of pitch speed control and autothrust thrust control correspond

- **a.** Altitude acquisition mode
- **b.** Level change mode
- **c.** Altitude hold mode

38. After a reversion during a climb, by which pitch movement is the commanded reference speed reached again?

- **a.** Pitch hold
- **b.** Pitch up
- **c.** Pitch down

39. A descent is flown with a very high negative vertical speed command and the aircraft approaches its upper speed limit. What does the autoflight system do?

- **a.** It transfers airspeed control to the pitch channel integrator
- **b.** It reduces the commanded vertical speed until the limit clears
- **c.** It reverts from vertical speed control to airspeed control

40. To what does the thrust rating taken up after a reversion in the descent correspond

- **a.** The minimum thrust previously set in vertical speed mode
- **b.** The maximum thrust previously set in vertical speed mode
- **c.** The thrust required for the commanded vertical speed

41. An excessive vertical speed causes an automatic reversion to level change. What is retained and what is modified?

- **a.** The flight path is retained and the speed target is modified
- **b.** The speed target is retained and the flight path is modified
- **c.** A new speed target is set and the flight path is retained

42. An aircraft is levelling off in altitude acquisition mode above the transition level. Which reference is the selected target expressed against? ●

- **a.** A radio height above ground
- **b.** A pressure altitude
- **c.** A QNH corrected altitude

43. The capture point is calculated from the vertical speed instead of being set at a fixed distance. Why is it done this way? ●

- **a.** So the altitude is reached without oscillating about it afterwards
- **b.** So the capture point can always be baro-corrected by the air data computer
- **c.** So the autothrust system can hold a constant airspeed during the capture

44. Two aircraft level off at the same altitude but close on it at different vertical speeds. Which one starts its capture further out? ●

- **a.** The one with the higher airspeed
- **b.** The one with the higher vertical speed
- **c.** The one with the lower vertical speed

45. As an aircraft in altitude acquisition gets closer and closer to the pre-selected altitude, what happens to its rate?

- **a.** It is held constant until the tolerance band
- **b.** It becomes steadily less
- **c.** It becomes steadily greater

46. In what time does the e-function representing the acquisition transition theoretically reach the final altitude

- **a.** The time to the capture point
- **b.** An infinite time
- **c.** The time to the tolerance band

47. The aircraft has just entered the tolerance band at the end of an altitude acquisition. Which mode becomes active and what is annunciated? ●

- **a.** Vertical speed, annunciated V/S = 0
- **b.** Altitude acquisition, annunciated ALT ACQ
- **c.** Altitude hold, annunciated ALT

48. On what condition does the comparator of the capture detector declare the altitude acquisition active ●

- **a.** Distance to go equals the calculated capture distance
- **b.** Distance to go exceeds the calculated capture distance
- **c.** Distance to go equals the width of the tolerance band

49. An aircraft is climbing in level change to a pre-selected flight level. Which mode is armed at that point?

- **a.** Altitude acquisition
- **b.** Altitude hold mode
- **c.** Speed and Mach control

50. During an altitude acquisition the autothrust holds a constant airspeed in speed mode. What is the effect of a pitch attitude change? ●

- **a.** The flight path angle changes, which modifies vertical speed
- **b.** The altitude error changes, which in turn modifies pitch attitude
- **c.** The airspeed changes, which in turn modifies vertical speed

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Answers — Level off, mode reversion and altitude acquire

19. **c** Levelling off means commanding no rate at all, so the reference becomes zero. (L1, PDF p. 162)

20. **b** The level off condition is annunciated as a commanded vertical speed of zero. (L1, PDF p. 162)

21. **b** Pre-selection arms the mode, so the armed annunciation appears first. (L1, PDF p. 166)

22. **a** The same controller drives the rate down to nothing and then keeps it there. (L2, PDF p. 162)

23. **a** The provisions vary so much from one type to another that no single description covers them all. (L2, PDF p. 164)

24. **b** The pitch channel holds the rate first, which is exactly why the speed can drift to a limit. (L2, PDF p. 164)

25. **c** The target is an altitude the pilot has set, read either against QNH or as a pressure altitude. (L2, PDF p. 166)

26. **a** Dialling the altitude is enough to start the detector running, long before any capture. (L2, PDF p. 166)

27. **c** The two quantities entering the calculation are the rate of closure and the height still to go. (L2, PDF p. 166)

28. **a** Below the transition altitude altitudes are flown on QNH, so the baro correction is applied. (L2, PDF p. 166)

29. **a** Reaching the point is the capture condition, so the mode goes from armed to active. (L2, PDF p. 166)

30. **a** Arming only starts the detector, and the loop is not closed until the capture condition is met. (L2, PDF p. 168)

31. **a** Height still to go is converted directly into a commanded rate, which is why the rate falls as the gap closes. (L2, PDF p. 168)

32. **a** An altitude error can only be the difference between the target altitude and the one being flown. (L2, PDF p. 168)

33. **a** The reference built from the height still to go is set against the rate the aircraft actually has, and the difference is the vertical speed error the controller works on. (L2, PDF p. 168)

34. **c** Level off changes nothing but the reference value: the same vertical speed controller works, now to a reference of zero. (L3, PDF p. 162)

35. **b** Envelope protection takes the rate target away from the pitch channel and gives it the speed target instead. (L3, PDF p. 164)

36. **c** The two channels swap jobs, so the autothrust gives up speed and goes to a fixed thrust rating. (L3, PDF p. 164)

37. **b** Pitch flying the speed with a fixed thrust rating is what defines level change. (L3, PDF p. 164)

38. **c** The aircraft was slow, so lowering the nose trades height for the speed it needs. (L3, PDF p. 164)

39. **c** The same envelope protection works at the top of the range as at the bottom, so vertical speed gives way to speed. (L3, PDF p. 164)

40. **a** The rating taken up on reversion in a descent is the minimum thrust the autothrust had set in vertical speed mode, while trying not to exceed the upper speed limit. (L3, PDF p. 164)

41. **b** Envelope protection keeps the speed and lets the profile change, which is the whole point of the reversion. (L3, PDF p. 165)

42. **b** Above the transition level flight levels are flown on standard setting, so the target is a pressure altitude. (L3, PDF p. 166)

43. **a** A high rate needs an earlier start, and scaling the point with rate is what stops the aircraft overshooting and hunting. (L3, PDF p. 166)

44. **b** Capture distance grows with vertical speed, so the faster approach needs the earlier start. (L3, PDF p. 166)

45. **b** The rate is held in proportion to the distance left, so as the distance shrinks the rate shrinks with it. (L3, PDF p. 166)

46. **b** An exponential approach never quite arrives, which is exactly why a tolerance band has to be defined. (L3, PDF p. 166)

47. **c** Acquisition hands over to hold at the edge of the band and the annunciation changes with it. (L3, PDF p. 166)

48. **a** Capture happens where the remaining distance has shrunk to the distance the detector has calculated. (L3, PDF p. 166)

49. **a** The acquisition is waiting for its capture condition, so it sits armed while the climb mode is active. (L3, PDF p. 167)

50. **a** With speed pinned by the autothrust, attitude translates straight into flight path angle and so into rate. (L3, PDF p. 168)

Altitude acquire in VNAV and altitude hold

PDF pages 170–174 · 29 questions

LEVEL 1 — RECALL

51. The flight management system is guiding the aircraft in the vertical navigation mode. In which phases of a flight can it do this? ●

- **a.** Climb, cruise and descend
- **b.** Take off, climb and cruise
- **c.** Cruise, descend and landing

52. At what deviation from the FM-altitude is profile or VNAV altitude hold entered automatically ●

- **a.** 20 ft
- **b.** 50 ft
- **c.** 100 ft

53. What does the ALT mode of the pitch channel maintain ●

- **a.** A pre-selected pitch attitude or trim setting
- **b.** A pre-selected rate or flight path angle
- **c.** A pre-selected altitude or flight level

54. What deviation from the pre-selected altitude do modern control laws permit ●

- **a.** 50 ft
- **b.** 100 ft
- **c.** 40 ft

LEVEL 2 — UNDERSTAND AND EXPLAIN

55. With which system does the profile or vertical navigation mode connect the autoflight system ●

- **a.** The inertial reference system
- **b.** The flight management system
- **c.** The air data computer

56. Over what span of a flight do the sequential segments of profile or vertical navigation guidance run ●

- **a.** From the capture point to the transition altitude
- **b.** From after take off to the instrument approach
- **c.** From the top of climb to the top of descent

57. How does the pilot give clearance for a flight level change demanded by the flight management system

- **a.** By selecting an appropriate altitude on the control panel
- **b.** By entering the new flight level into the flight management system
- **c.** By pressing the profile button at the autoflight control panel

58. An aircraft in profile mode approaches the FM-altitude within fifty feet. What happens? ●

- **a.** The system automatically transits to altitude hold
- **b.** The system automatically reverts to vertical speed
- **c.** The system automatically arms altitude acquisition

59. In which mode does the autothrust system work while the altitude hold mode is active ●

- **a.** Speed mode
- **b.** Thrust mode
- **c.** Idle mode

60. The altitude hold button is pressed while the aircraft is climbing. Which altitude does the system then maintain?

- **a.** The altitude momentarily reached
- **b.** The pre-selected altitude on the panel
- **c.** The altitude at which the climb began

61. The crew has not pressed the altitude hold button. Under what condition does the altitude hold mode still come in automatically? ●

- **a.** On passing the transition altitude during a climb or descent
- **b.** On approaching a pre-selected altitude with acquisition armed
- **c.** On approaching a pre-selected altitude with acquisition active

62. Why is the soft altitude mode not explicitly annunciated ●

- **a.** It is subordinate to the altitude hold mode
- **b.** It is a function of the altitude memory alone
- **c.** It is an armed mode only and never an active one

63. Below what size of airspeed disturbance is the soft altitude deviation permitted ●

- **a.** 10 Kts
- **b.** 5 Kts
- **c.** 4 Kts

LEVEL 3 — APPLY AND REASON

64. The flight management system requests a climb. Why must the autoflight system have a pilot input before it starts that climb? ●

- **a.** To prevent altitude changes being made without ATC-clearance
- **b.** To prevent the autothrust system reverting to thrust mode
- **c.** To prevent the flight management system losing its reference

65. Where must both the panel selection and the FM-altitude lie before a climb to an FM-altitude is cleared ●

- **a.** Below the momentary altitude of the aircraft
- **b.** Exactly at the momentary altitude of the aircraft
- **c.** Above the momentary altitude of the aircraft

66. How must the FM-altitude stand against the panel selection for a climb to be cleared ●

- **a.** Exactly equal to it
- **b.** Equal to or above it
- **c.** Equal to or below it

67. Where must both the panel selection and the FM-altitude lie before a descent to an FM-altitude is cleared ●

- **a.** Exactly at the momentary altitude of the aircraft
- **b.** Below the momentary altitude of the aircraft
- **c.** Above the momentary altitude of the aircraft

68. A descent to an FM-altitude is to be cleared. How must the FM-altitude stand against the panel selection? ●

- **a.** Exactly equal to it
- **b.** Equal to or below it
- **c.** Equal to or above it

69. An aircraft climbs in profile mode with the panel selection above the level the flight management system is aiming for. Where does it level off?

- **a.** At the selected flight level
- **b.** At the FMS target level
- **c.** Midway between the two levels

70. An aircraft climbs in profile mode with the panel altitude below the FMS target level. Where does it level off?

- **a.** It continues the climb unrestricted
- **b.** At the flight management target level
- **c.** At the selected flight level

71. An aircraft descends in profile mode with the panel altitude above the FMS target level. Where does it level off?

- **a.** At the selected flight level
- **b.** At the flight management target level
- **c.** At the transition level

72. The altitude hold button is pressed during a descent. Why does the aircraft then have to acquire that altitude again?

- **a.** Because it undershoots that altitude while still descending
- **b.** Because the autothrust reverts to thrust mode and sets flight idle
- **c.** Because the altitude memory is only synchronised once the button is released

73. In which condition is manual activation of the altitude hold mode mainly used

- **a.** When the aircraft is approaching a pre-selected altitude
- **b.** When the aircraft is in the capture phase
- **c.** When the aircraft is already in level flight

74. The soft altitude mode lets the aircraft wander within its band. What is traded for what? ●

- **a.** Potential energy as airspeed against kinetic energy as altitude
- **b.** Kinetic energy as thrust against potential energy as airspeed
- **c.** Potential energy as altitude against kinetic energy as airspeed

75. What is the immediate benefit of the energy trade allowed in the soft altitude mode ●

- **a.** Fewer corrections of the altitude reference in memory
- **b.** Fewer transient oscillations about the altitude
- **c.** Fewer changes of engine thrust for airspeed control

76. Which two results does the smoother thrust control of the soft altitude mode bring? ●

- **a.** Reduced fuel consumption and increased passenger comfort
- **b.** Reduced engine wear and increased thrust response rate
- **c.** Reduced altitude deviation and increased airspeed stability

77. The altitude hold mode is not yet active. What is the altitude memory doing?

- **a.** It is holding the last altitude at which the mode was active
- **b.** It is updated permanently with the pre-selected altitude from the panel
- **c.** It is updated permanently with the current air data altitude

78. After the flight path angle has changed in altitude hold, what is the order of the remaining effects

- **a.** Vertical speed changes and then altitude changes
- **b.** Thrust changes and then altitude changes
- **c.** Altitude changes and then vertical speed changes

79. The altitude hold loop is compared with the altitude acquisition cascade. Why does altitude hold need no intermediate rate command?

- **a.** Because its reference value is not stored in any memory at all
- **b.** Because its vertical speed is supplied by the autothrust system
- **c.** Because its own controller sets the attitude reference directly

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Answers — Altitude acquire in VNAV and altitude hold

51. **a** The three named phases are climb, cruise and descend. (L1, PDF p. 170)

52. **b** The altitude band for this transit is fifty feet. (L1, PDF p. 170)

53. **c** The mode is named for what it holds, which is the altitude or flight level already selected. (L1, PDF p. 172)

54. **b** The printed allowance in the soft altitude mode is a hundred feet either side. (L1, PDF p. 172)

55. **b** Vertical navigation is flown on data the flight management system supplies, so the two are coupled. (L2, PDF p. 170)

56. **b** The sequential segments are stated as running from just after take off through to the approach. (L2, PDF p. 170)

57. **a** The altitude selector on the panel is the place where the crew confirm the clearance to the system. (L2, PDF p. 170)

58. **a** Fifty feet of deviation is the printed trigger for the automatic change into hold. (L2, PDF p. 170)

59. **a** With the pitch channel holding height, the autothrust takes the speed and works to a speed target. (L2, PDF p. 172)

60. **a** A manual selection freezes whatever altitude the aircraft has at that instant. (L2, PDF p. 172)

61. **c** The acquisition must already be running, not merely armed, for the hold to follow automatically. (L2, PDF p. 172)

62. **a** It works underneath altitude hold rather than beside it, so it gets no annunciation of its own. (L2, PDF p. 172)

63. **c** The allowance applies only to disturbances below four knots. (L2, PDF p. 172)

64. **a** The pilot input stands in for the clearance, so the aircraft never changes level on its own. (L3, PDF p. 170)

65. **c** Both targets have to lie above the aircraft before a climb can be cleared. (L3, PDF p. 170)

66. **c** In a climb the panel selection is the ceiling the crew have cleared, so the flight management target may not sit above it. (L3, PDF p. 170)

67. **b** Both targets have to lie below the aircraft before a descent can be cleared. (L3, PDF p. 170)

68. **c** In a descent the panel selection is the floor the crew have cleared, so the flight management target may not sit below it. (L3, PDF p. 170)

69. **b** In a climb the aircraft stops at whichever of the two lies lower, which here is the flight management target. (L3, PDF p. 171)

70. **c** The crew input acts as the cleared ceiling, so the aircraft stops there. (L3, PDF p. 171)

71. **a** In a descent the aircraft stops at whichever of the two lies higher, which here is the crew input. (L3, PDF p. 171)

72. **a** The altitude is frozen while the aircraft is still going down, so it undershoots and then has to acquire that altitude again. (L3, PDF p. 172)

73. **c** If there is no rate to arrest there is nothing to overshoot, so the manual selection is clean. (L3, PDF p. 172)

74. **c** Height is the potential energy and speed the kinetic energy, and the band lets one be swapped for the other. (L3, PDF p. 172)

75. **c** Letting the altitude move absorbs the speed disturbance so the engines do not have to. (L3, PDF p. 172)

76. **a** The two stated results of the smoother thrust are less fuel burned and a smoother ride. (L3, PDF p. 172)

77. **c** Continuous updating keeps the stored reference equal to the actual altitude so there is no step at engagement. (L3, PDF p. 174)

78. **a** Rate comes before height, because altitude only moves once a rate exists. (L3, PDF p. 174)

79. **c** Holding needs no rate step because the altitude controller produces the attitude reference itself. (L3, PDF p. 174)

VNAV and LNAV modes, VOR navigation

PDF pages 176–188 · 48 questions

LEVEL 1 — RECALL

80. What relation do the several operating modes covered by the generic vertical navigation term bear to one another ●

a. They have separate annunciations
b. They have identical meaning
c. They have opposing control laws

81. A crew uses the term vertical navigation. Which mode name has the same meaning? ●

a. Altitude acquisition
b. Level off
c. Profile

82. Which phase follows the arm phase of the horizontal navigation mode ●

a. The over station phase
b. The active phase
c. The on course phase

83. Above which radio height can the autopilot and flight director be coupled with the flight management system ●

a. 50 ft
b. 100 ft
c. 30 ft

84. How is a VOR radial described ●

a. A variable line of position
b. A fixed line of position
c. A fixed line of equal altitude

85. How many sub-modes does the VOR controller have to distinguish ●

a. Two
b. Four
c. Three

86. The VOR mode has just been selected but the angular deviation limit has not yet been passed. What does the flight mode annunciator show? ●

a. VOR capture phase
b. VOR mode active
c. VOR mode armed

87. Which pair of values does the roll controller use as the criterion for transiting to the On-Course Phase ●

a. VOR deviation 15° and heading error 5°
b. VOR deviation 5° and heading error 15°
c. VOR deviation 2° and heading error 10°

88. A steady heading error is sitting at the input of the washout — roughly which time constant governs how quickly it decays? ●

a. 10 sec
b. 100 sec
c. 200 sec

89. Which criterion is used to recognise the cone of confusion ●

- **a.** A deviation signal change of more than 5 mV/sec or more than 0,8°/sec
- **b.** A deviation signal change of more than 8 mV/sec or more than 0,2°/sec
- **c.** A deviation signal change of more than 8 mV/sec or more than 0,5°/sec

LEVEL 2 — UNDERSTAND AND EXPLAIN

90. Which four sub-modes does managed level-change comprise ●

- **a.** Climb, altitude hold, altitude constraint, descent
- **b.** Climb, altitude acquisition, level off and descent
- **c.** Climb, cruise, instrument approach and descent

91. Which two systems are coupled with the flight management system in all the vertical navigation modes ●

- **a.** The autopilot/director and the autothrust system
- **b.** The air data computer and the inertial reference system
- **c.** The roll channel of the autopilot and the yaw damper

92. In the vertical navigation modes, which channel performs the vertical guidance? ●

- **a.** The pitch channel
- **b.** The autothrust channel
- **c.** The roll channel

93. Which unit determines the speed and Mach reference values in the vertical navigation modes ●

- **a.** The flight management system
- **b.** The autothrust control system
- **c.** The central air data computer

94. Where the autothrust system is working in thrust mode, what decides the source of the thrust ratings it uses? ●

- **a.** The type of aircraft
- **b.** The selected altitude
- **c.** The momentary vertical speed

95. From the drift between the wanted on-track point and what else does the pilot work out a wind correction angle

- **a.** The actual position
- **b.** The desired track angle
- **c.** The selected course

96. The capture phase of the navigation mode has just been entered. What is annunciated? ●

- **a.** NAV active
- **b.** NAV armed
- **c.** Heading select active

97. Which unit directs the autoflight system to intercept and follow the track once the navigation mode is active ●

- **a.** The flight management system
- **b.** The capture detector
- **c.** The inertial navigation system

98. With the help of which autopilot channel does the VOR mode intercept and follow a radial ●

a. The roll channel
b. The pitch channel
c. The yaw channel

99. Besides selecting an appropriate receiver frequency, what else is required before the VOR mode can be activated? ●

a. The intercept angle must be determined by selecting a track
b. The radial must be determined by selecting a course
c. The radial must be determined by selecting a heading

100. In which order do the three VOR sub-modes follow the flight phase sequence ●

a. Capture, on course and over station
b. Capture, over station and on course
c. Arm, capture and over station

101. The VOR mode has just entered its capture phase. What is that phase intended to achieve? ●

a. Guidance of the aircraft through the area where the deviation is useless
b. Guidance of the aircraft along the radial with cross wind correction
c. Transfer from an intercept heading onto the radial without oscillations

102. Besides guiding the aircraft on the radial from or to the station, what does the on course phase do ●

a. It corrects any cross wind disturbances
b. It corrects any deviation rate above the threshold
c. It corrects the pre-selected course for station passage

103. At approximately what angular deviation is the transfer to the pre-selected course said to start ●

a. 5 degrees
b. 2 degrees
c. 10 degrees

104. Which unit is activated first on selection of the VOR mode to measure angular deviation from the pre-selected course ●

a. The lateral beam sensor
b. The capture detector
c. The course comparator

105. What becomes the controlled value of the third loop that augments heading control during the capture phase ●

a. The pre-selected course value
b. The actual aircraft heading
c. The actual VOR-radial

106. With what is the pre-selected course compared to obtain the deviation used by the outermost VOR loop

a. The actual roll attitude from the gyros
b. The actual heading from the compass system
c. The actual radial at the aircraft's position

107. Why does the roll controller change over to the On-Course Phase once the preselected radial is almost reached ●

a. To hand the lateral guidance over to the heading hold controller
b. To increase the rate at which the heading error is reduced
c. To improve its behaviour with respect to cross wind disturbances

108. An aircraft tracking a VOR radial meets a permanent cross wind component — what does comparing the VOR deviation with the heading error in S3 then produce? ●

a. A permanent oscillation about the radial
b. A permanent loss of the VOR deviation signal
c. A permanent Stand-Off from the radial

109. Which heading error does the roll controller have to leave standing while an aircraft tracks a radial in a steady crosswind ●

a. The one equivalent to the preselected course
b. The one equivalent to the wind correction angle
c. The one equivalent to the VOR course deviation angle

110. An aircraft is established on a pre-selected radial in a steady crosswind — how does it travel along that radial? ●

a. Wings level, with the heading error equal to the wind correction angle
b. In small banked corrections, with the heading error slowly decaying
c. Wings level, with the heading error equal to the VOR deviation angle rate

111. An aircraft is passing the cone of confusion overhead a VOR — for how long after leaving that area does the deviation signal stay switched off? ●

a. A further 100 seconds
b. A further 20 seconds
c. A further 10 seconds

LEVEL 3 — APPLY AND REASON

112. What determines the operating mode of the autothrust system in the vertical navigation modes ●

a. The roll channel
b. The air data computer
c. The pitch channel

113. A waypoint has to be passed at a given altitude with a given airspeed maintained during a climb. What does the flight management computer calculate? ●

a. An optimum flight path angle for the required thrust
b. An optimum vertical speed for the required altitude
c. An optimum thrust for the required flight path angle

114. By selecting what does the flight management system determine the control loops for the different flight phases

a. Appropriate thrust ratings for the autothrust system
b. Appropriate pitch channel modes of the autopilot
c. Appropriate reference altitudes for the altitude memory

115. A fixed inertial flight path is being flown in heading select mode. Why must the heading reference be corrected from time to time?

a. Because the required value depends on the prevailing wind
b. Because the compass system drifts away from magnetic north
c. Because the roll channel loses its attitude reference in a turn

116. Besides an inertial system or a flight management system with position sensors, what third source makes automatic lateral guidance possible ●

a. A compass system with a heading hold function
b. An air data computer with altitude correction
c. A radio navigation system with track capability

117. An active flight plan was entered into the flight management system during cockpit preparation. How will the autopilot couple to it? ●

a. Manually only, after take off
b. Only once the capture phase begins
c. Automatically, without crew action

118. The navigation mode is active. Apart from pressing its button again, how is it terminated? ●

a. Automatically, as soon as another vertical mode is selected
b. Automatically, as soon as another lateral mode is selected
c. Automatically, as soon as the cross-track distance grows again

119. The navigation mode is active and a heading change is needed. How is that change actually brought about?

a. By track error values fed to the roll channel inner loop
b. By roll attitude reference values fed to the roll inner loop
c. By heading reference values fed to the roll channel outer loop

120. An aircraft tracking a radial passes overhead the VOR station. Why does the VOR mode need a separate over station phase? ●

a. Because the deviation signal is useless in the cone of confusion
b. Because the receiver frequency must be re-tuned as the station is passed
c. Because the pre-selected course must be reversed as the station is passed

121. The radial deviation rate rises above eight millivolts per second. What does the roll controller do? ●

a. It transits to heading control and maintains the present heading
b. It transits to capture control and re-intercepts the radial
c. It transits to heading control and reverses the pre-selected course

122. An aircraft is closing on the pre-selected course. Why must the transfer onto it start early rather than at the course itself?

a. So the lateral beam sensor has time to measure the deviation
b. So the flight mode annunciator can indicate the mode armed
c. So the bracketing is soft and free of oscillations

123. The aircraft passes the angular deviation limit on its way to the radial. What does the lateral beam sensor do? ●

a. It switches the roll controller into heading control
b. It switches the roll controller into the on course phase
c. It switches the roll controller into the capture phase

124. A signal used by the VOR mode becomes invalid in flight. What actually happens, given the permanent monitoring of signal validity?

- **a.** The VOR mode goes back to capture
- **b.** The VOR mode holds its last command
- **c.** The VOR mode is terminated

125. A washout has been added to the navigation loop and the aircraft is tracking a radial in a steady crosswind — what is the relationship between the aircraft heading and the PSC? ●

- **a.** The heading differs from the PSC by the measured VOR deviation angle
- **b.** The heading differs from the PSC by the wind correction angle
- **c.** The heading is aligned with the PSC and the track is offset

126. How does the washout let the VOR deviation prevail over the heading error in the long run ●

- **a.** Only changes of VOR deviation are still compared with the heading error
- **b.** Both signals are compared but the heading error is given a lower gain
- **c.** Only changes of heading error are still compared with the VOR deviation

127. The VOR deviation signal has been switched off and the heading error signal connected to S3 during the Over Station Phase — how does the controller then behave? ●

- **a.** The heading loop is off and it acts as a navigation controller only
- **b.** The navigation loop is off and it acts as a roll rate controller only
- **c.** The navigation loop is off and it acts as a heading controller only

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Answers — VNAV and LNAV modes, VOR navigation

80. **b** The listed names are different labels for the same thing. (L1, PDF p. 176)

81. **c** Profile is one of the names printed for the same mode as vertical navigation; altitude acquisition and level off are separate pitch modes. (L1, PDF p. 176)

82. **b** This mode runs as an arm phase followed by an active phase; on course and over station phases belong to the VOR mode. (L1, PDF p. 178)

83. **c** The printed minimum radio height for coupling is thirty feet. (L1, PDF p. 178)

84. **b** The radial is a line of position fixed in space relative to the station. (L1, PDF p. 182)

85. **c** Capture, on course and over station make three. (L1, PDF p. 182)

86. **c** Until the limit is passed the mode is only waiting, so it reads armed. (L1, PDF p. 184)

87. **b** The printed criterion is a VOR deviation of 5° or less together with a heading error of 15° or less. (L1, PDF p. 186)

88. **b** The washout time constant is approximately 100 sec. (L1, PDF p. 188)

89. **c** Those are the two printed rates that define the cone of confusion. (L1, PDF p. 188)

90. **a** Climb, altitude hold, altitude constraint and descent are the four sub-modes of managed level-change. (L2, PDF p. 176)

91. **a** Both the guidance and the thrust systems have to be tied to the flight management system for automatic vertical guidance. (L2, PDF p. 176)

92. **a** Vertical guidance is a pitch task and thrust control is left to the autothrust. (L2, PDF p. 176)

93. **a** The targets come from the flight management system, which is what makes the guidance managed. (L2, PDF p. 176)

94. **a** The source of the ratings varies from one aircraft type to another. (L2, PDF p. 176)

95. **a** Drift is measured between where the aircraft should be and where it actually is. (L2, PDF p. 178)

96. **a** Entering the capture phase makes the mode active, and the annunciation says so. (L2, PDF p. 178)

97. **a** The flight management system holds the flight plan, so it is what directs the interception. (L2, PDF p. 178)

98. **a** A radial is a lateral path, so the roll channel flies it. (L2, PDF p. 182)

99. **b** The radial is told to the system as a course set on the pre-selected course knob. (L2, PDF p. 182)

100. **a** The aircraft joins the radial, tracks it, and only then passes the station. (L2, PDF p. 182)

101. c Capture is the joining manoeuvre, and it must settle the aircraft on the radial cleanly. (L2, PDF p. 182)

102. a Tracking a radial means holding it against the wind, which is what this phase does. (L2, PDF p. 182)

103. b The capture starts early, at about two degrees of angular deviation. (L2, PDF p. 184)

104. a The lateral beam sensor is the device that watches the angular deviation while the mode is armed. (L2, PDF p. 184)

105. c The outermost loop added at capture controls the radial the aircraft is actually on. (L2, PDF p. 184)

106. c Deviation is the difference between the course wanted and the radial the aircraft is actually on. (L2, PDF p. 184)

107. c Cross wind disturbance behaviour is what the change of phase improves. (L2, PDF p. 186)

108. c A standing cross wind leaves a residual difference in S3, and that residual is the permanent Stand-Off. (L2, PDF p. 186)

109. b The crab angle must be kept, so the heading error equal to the wind correction angle is left alone. (L2, PDF p. 186)

110. a Once the washout has settled the crab is held and no roll command remains, so the aircraft drifts wings level. (L2, PDF p. 188)

111. b The printed period is the cone of confusion plus a further 20 seconds. (L2, PDF p. 188)

112. c The pitch channel takes the lead and the autothrust mode follows from what pitch is doing. (L3, PDF p. 176)

113. c The constraint fixes the path to be flown, so what the computer works out is the thrust that path needs. (L3, PDF p. 176)

114. b The flight management system works by picking pitch channel modes, and those modes define the loops. (L3, PDF p. 176)

115. a A heading only produces the wanted track for one particular wind, so it has to be reset as the wind changes. (L3, PDF p. 178)

116. c The third option listed is a radio aid that defines a track, such as a radial or a localizer. (L3, PDF p. 178)

117. c An active flight plan already loaded before departure gives automatic coupling. (L3, PDF p. 178)

118. b Two lateral modes cannot run together, so selecting another one ends this one. (L3, PDF p. 178)

119. b The aircraft is turned by banking it, so the command reaching the inner loop is a roll attitude. (L3, PDF p. 180)

120. a With no usable deviation overhead, the controller cannot keep steering on it and needs another rule. (L3, PDF p. 182)

121. a A fast moving deviation signals the cone of confusion, so the controller freezes the heading and flies on. (L3, PDF p. 182)

122. c Turning early gives room for a gentle turn that settles on the course instead of swinging through it. (L3, PDF p. 184)

123. c The sensor passing its limit is the capture trigger, and it works through the operating mode logic. (L3, PDF p. 184)

124. c Monitoring drops the mode rather than letting it fly on a missing signal. (L3, PDF p. 184)

125. b With the washout fitted the radial is tracked, so the crab is carried as a heading difference from the PSC. (L3, PDF p. 186)

126. c The washout passes only the changing part of the heading error, so the steady deviation signal dominates. (L3, PDF p. 186)

127. c With the deviation signal gone and the washout bypassed, only the heading loop remains in use. (L3, PDF p. 188)

Localizer and back beam modes

PDF pages 190–198 · 26 questions

LEVEL 1 — RECALL

128. Which kind of approach does the localizer mode support when it is used as an independent auto flight mode ●

a. A Precision Approach
b. A Non-Precision Approach
c. A fully automatic landing

129. Which flight mode annunciation belongs to the localizer Capture Phase ●

a. LOC (TRACK)
b. LOC (ARM)
c. LOC (ACTIVE)

130. Which other name is given to the back beam of a localizer ●

a. The back radial
b. The reciprocal localizer
c. The back course

131. Which two forms of guidance can the back beam mode provide ●

a. Autopilot guidance or radar vectoring
b. Flight director guidance or auto-thrust guidance
c. Flight director guidance or autopilot guidance

132. A back beam approach is being set up on the HSI — by how much must the deviation scale and the deviation bar be rotated? ●

a. 360°
b. 180°
c. 90°

133. How many radials does a VOR provide compared with the number of beams a localizer transmitter generates ●

a. 36 radials against one beam
b. 360 radials against one beam
c. 180 radials against one beam

LEVEL 2 — UNDERSTAND AND EXPLAIN

134. A crew selects Approach/Land on the autoflight control panel — what happens to the localizer mode? ●

a. It must still be armed separately by its own key
b. It is activated automatically as a submode
c. It is inhibited until the glide path is captured

135. The lateral annunciation changes from armed to active — which phase has just begun? ●

a. The Capture Phase
b. The Loc-Track Phase
c. The Approach-Track Phase

136. What are angular deviations converted into during the Approach-Track Phase, and why ●

a. Into roll rate commands, to counteract the funnel effect
b. Into lateral distances, to counteract the beam convergence effect
c. Into lateral distances, to counteract the funnel effect

137. From which two quantities does the Capture Detector calculate the Capture Point ●

a. The heading error and the LOC deviation rate
b. The heading error and the LOC deviation angle
c. The roll attitude and the LOC deviation rate

138. Why is the capture threshold of the localizer mode called adaptive ●

a. Because the capture point is reset each time the deviation changes sign
b. Because the capture point location depends particularly on the deviation rate
c. Because the capture point location depends particularly on the radio altitude

139. Which property of the aircraft are the Approach-Track control laws written to exploit ●

a. Its weather cock stability
b. Its spiral stability
c. Its longitudinal stability

140. Why does the back beam mode count as a Non-Precision Approach ●

a. Because no vertical guidance is available in this mode
b. Because no lateral guidance is available in this mode
c. Because the deviation signal polarity is inverted in this mode

141. A crew is setting up for a localizer back beam approach — which course must be set as the pre-selected course? ●

a. The back course
b. The runway heading plus 90°
c. The front course

142. The deviation bar on the HSI has to read correctly during a back beam approach — how is the required 180° rotation obtained? ●

a. By inverting the heading signal from the compass system
b. By setting the back course as the pre-selected course
c. By setting the front course as the pre-selected course

143. What happens to the polarity of the localizer deviation signal on the back beam ●

a. It is inverted and the inversion has to be undone in the navigation loop
b. It is inverted and the inversion is left in place for the navigation loop
c. It is unchanged and the heading error is inverted instead

144. Which inverter reverses the localizer deviation signal for a back beam approach ●

a. Inverter 2
b. Inverter 1
c. The rate taker

145. Which two signals are added to form the roll attitude reference value in the heading loop

a. The heading error and the localizer deviation
b. The heading error and the roll attitude error
c. The localizer deviation and the roll rate

LEVEL 3 — APPLY AND REASON

146. The localizer mode has been selected on its own on the autoflight control panel as an independent lateral mode — which phase is left out? ●

a. The Approach-Track Phase
b. The Localizer-Track Phase
c. The Capture Phase

147. Two aircraft intercept the same localizer from very different distances to the transmitter antenna — what does the adaptive capture threshold guarantee? ●

a. The best possible beginning of the transition onto the beam for both
b. The same bank angle throughout the transition onto the beam for both
c. The same intercept angle onto the beam for both

148. The localizer deviation signal has just fallen to 60 mV during an intercept — what is now fed into the outer control loop?

a. The PSC error, through a wash out
b. The PSC error, directly without a wash out
c. The localizer deviation rate, through a wash out

149. The aircraft is in the Approach-Track Phase and a gust pushes the heading off — under what condition will the roll controller still act?

a. If the gust changes the sink rate or displaces it from the beam
b. If the gust changes the bank angle or displaces it from the beam
c. If the gust changes the bank angle or the heading error exceeds 0,5°

150. Why does a further 180° have to be subtracted from the heading error built on the preselected front course ●

a. Because the actual heading error has to be taken from the back course
b. Because the actual heading error has to be taken from the front course
c. Because the deviation bar polarity has already been reversed

151. What is the inversion applied to the heading error of a back beam approach equivalent to

a. Subtraction of 180° from the localizer deviation obtained at S1
b. Subtraction of 180° from the heading error obtained at S1
c. Addition of 180° to the heading error obtained at S2

152. The pilot is flying the back beam manually on the flight director and keeps the roll bar centred — what is the result?

a. There is no heading error and the back beam is tracked
b. There is no localizer deviation and the roll bar stays centred
c. There is no roll attitude error and the back beam is tracked

153. How does the sequence of effects of the flight director differ from that of the autopilot on a back beam approach

a. The pilot is directly involved as part of the inner control loop
b. The inner control loop is replaced by the roll attitude limitation
c. The pilot is directly involved as part of the navigation loop

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Answers — Localizer and back beam modes

128. b Used on its own the localizer mode gives lateral guidance only, which is a Non-Precision Approach. (L1, PDF p. 190)

129. b The Capture Phase is annunciated as LOC (ARM). (L1, PDF p. 190)

130. c Back course is the alternative name for the back beam. (L1, PDF p. 194)

131. c The mode works for either the flight director or the autopilot. (L1, PDF p. 194)

132. b The printed rotation for a correct back beam indication is 180°. (L1, PDF p. 196)

133. b A VOR provides 360 radials, whereas the localizer transmitter provides a single beam. (L1, PDF p. 198)

134. b As a submode of Approach/Land it comes up with that selection and needs no separate key. (L2, PDF p. 190)

135. b That annunciation change marks the beginning of the Loc-Track Phase. (L2, PDF p. 190)

136. c They become lateral distances, which is what counteracts the funnel effect. (L2, PDF p. 190)

137. a Heading error and rate of change of localizer deviation are the two printed inputs. (L2, PDF p. 192)

138. b The threshold moves with the closure rate, which is why it is called adaptive. (L2, PDF p. 192)

139. a Weather cock stability is the printed property relied on near the ground. (L2, PDF p. 192)

140. a Only lateral guidance exists on the back beam, so the approach is non-precision. (L2, PDF p. 194)

141. c The front course, not the back course, is what the system expects at the PSC. (L2, PDF p. 194)

142. c Selecting the front course at the PSC rotates the scale and bar by 180°. (L2, PDF p. 196)

143. a The mirror image situation inverts the signal, and the controller has to reverse that inversion. (L2, PDF p. 196)

144. a The deviation signal is reversed by Inverter 2, while Inverter 1 works on the heading error. (L2, PDF p. 198)

145. a The reference roll attitude comes from the sum of heading error and localizer deviation formed in S2. (L2, PDF p. 198)

146. a Selecting LOC as an independent lateral mode omits the Approach-Track Phase. (L3, PDF p. 190)

147. a The threshold is written to be independent of range, so the turn-on point is right in either case. (L3, PDF p. 192)

148. a Reaching that beam value starts the Loc-Track Phase, in which the PSC error enters through the wash out. (L3, PDF p. 192)

149. b Only upsets that move the bank angle or move the aircraft off the beam are corrected. (L3, PDF p. 192)

150. a The usable heading error is referenced to the back course, so another 180° comes off. (L3, PDF p. 196)

151. b Inverter 1 reverses the sign, which amounts to taking 180° off the S1 heading error. (L3, PDF p. 198)

152. c Centring the bar removes the roll attitude error, which is what the flight director displays. (L3, PDF p. 198)

153. a With the flight director the pilot closes the inner loop himself. (L3, PDF p. 198)

Approach and land mode, glideslope submode

PDF pages 200–208 · 33 questions

LEVEL 1 — RECALL

154. Which common mode is available for the flight director only ●

a. TAKE OFF
b. ILS APPROACH/LAND
c. GO AROUND

155. Below which height above ground can the Approach/Land or Go Around mode no longer be deactivated by pressing its button again ●

a. 400 ft
b. 1500 ft
c. 1000 ft

156. Below which height above ground can the Approach or Land mode be left in only two ways ●

a. 1500 ft
b. 500 ft
c. 1000 ft

157. Which vertical flight mode annunciation appears once the prerequisites for the glide path mode are met ●

a. G/S - CAPTURE
b. G/P - ARM
c. G/P - ACTIVE

158. Within which angular range is the glide path signal usable ●

a. +/- 0,5°
b. +/- 1,5°
c. +/- 1,0°

159. Below which height AGL does beam convergence make direct glide path control by zeroing the angular deviation unworkable ●

a. 500 ft
b. 1500 ft
c. 1000 ft

160. Which unit monitors the angular glide path deviation signal as soon as Approach/Land is selected ●

a. The over station sensor
b. The vertical beam sensor
c. The capture point detector

161. Below which angular deviation does the vertical beam sensor trigger the glide path capture phase ●

a. 2°
b. 0,5°
c. 0,2°

LEVEL 2 — UNDERSTAND AND EXPLAIN

162. An aircraft is at 300 ft AGL with the Approach/Land mode engaged and the crew presses the Approach button again — what happens? ●

a. The mode disengages immediately
b. The mode stays in operation
c. The mode reverts to the localizer submode

163. What does a precision approach require on board the aircraft or on the ground

a. A localizer transmitter showing deviation from the runway centre line
b. A glide path indicator showing deviation from the intended glide path
c. A marker beacon showing the distance to the runway threshold

164. What do all non-precision approaches have in common ●

a. They lack any lateral guidance
b. They lack any distance information
c. They lack any vertical guidance

165. The pilot decides to discontinue an approach — according to what must the missed approach be flown? ●

a. The procedures published in instrument approach charts or ATC guidance
b. The procedures published in the aircraft flight manual or ATC guidance
c. The procedures stored in the flight management system or the approach plate

166. How are the terms Approach mode and Land mode related

a. The approach mode is armed and the land mode is active
b. They are equivalent names for the same mode
c. The land mode is a sub-mode of the approach mode

167. An automatic approach is being flown with the Approach mode engaged — which span of flight phases does it control? ●

a. From glide path intercept, via flare, until the end of the missed approach
b. From localizer intercept, via flare, until the take off mode is selected
c. From localizer intercept, via glide path intercept, until roll out

168. An aircraft is below the JAR-AWO height with the Land mode engaged — what are the only two ways of deactivating it? ●

a. Selecting Go Around or selecting another vertical mode
b. Selecting Go Around or turning off the autopilot
c. Turning off the autopilot or turning off the auto-thrust

169. Which two settings are a prerequisite before the glide path mode can work ●

a. The back course of the active runway and the runway heading at the PSC
b. The ILS frequency of the active runway and the decision height at the PSC
c. The ILS frequency of the active runway and the runway heading at the PSC

170. The aircraft is in altitude hold on the localizer and passes an angular deviation of approximately -0,2° — what happens? ●

a. Altitude hold switches off and a descent starts
b. Altitude hold is retained until the beam centre is reached
c. Altitude hold switches off and the flare law is armed

171. What does an angular deviation of approximately -0,2° correspond to on the glide path deviation indicator ●

a. Roughly one dot below
b. Roughly half a dot above
c. Roughly half a dot below

172. An aircraft on an automatic approach descends through approximately 70 ft AGL — why is glide path control terminated at the latest here? ●

a. The glide path signal is outside its usable angular range
b. The radio altimeter becomes unreliable through ground reflections now
c. The glide path signal becomes unreliable through ground reflection

173. An aircraft holds a constant vertical distance from the glide path as it closes on the transmitter — what happens to the angular deviation? ●

a. It stays constant
b. It decreases permanently
c. It increases permanently

174. Which name is given to the permanently increasing loop amplification of a closing glide path ●

a. Beam convergence
b. Beam sensitivity
c. Funnel effect

175. Which two quantities does the first estimation method use to estimate the vertical glide path deviation ●

a. Angular deviation and ground speed
b. Angular deviation and radio height
c. Radio height and ground speed

176. Which quantity is fed back to form the inner loop of the glide slope mode ●

a. The pitch attitude
b. The radio height
c. The vertical speed

177. Why will an aircraft hold a constant sink rate during a steady descent without controller action ●

a. Because of the pitch trim integrator
b. Because of its inherent stability
c. Because of its weather cock stability

LEVEL 3 — APPLY AND REASON

178. The Approach/Land mode is engaged at 200 ft AGL and the crew decides to discontinue the approach — which mode can still take over? ●

a. The heading select mode
b. The Go Around mode
c. The altitude hold mode

179. An automatic landing is being flown and the glide path has just been left — what is held until the flare phase begins? ●

a. The momentary pitch attitude
b. The momentary sink rate
c. The momentary glide slope angle

180. An autopilot is certified for category II approaches only — what must happen at 100 ft radio height? ●

a. It must be disengaged and the flare flown by the flight director
b. It must be switched to the flare sub-mode for the landing
c. It must be disengaged and the landing flown manually

181. Beam convergence has been left uncompensated on an ILS approach — how does the glide path controller behave at long range and close in? ●

a. Sluggish far out and unable to hold the beam near the runway
b. Prone to oscillation far out and sluggish near the runway
c. Sluggish far out and prone to oscillation near the runway

182. Why do estimation methods have to be used to compensate the apparent increase in amplification ●

a. The vertical deviation from the G/P beam is generally not available
b. The angular deviation from the G/P is generally not available
c. The slant range to the G/P-transmitter is generally not available

183. The second estimation method is in use and the aircraft passes the outer marker — what is started there, and what has to be known? ●

a. A time proportional estimation, with the height of the aircraft over the marker
b. A radio height proportional estimation, with the distance from OM to the threshold
c. A time proportional estimation, with the distance from OM to the transmitter

184. A glide slope deviation appears while an aircraft is in a steady descent — which loops detect it and act on it? ●

a. The pitch outer loop detects it and the pitch inner loop changes the sink rate
b. The pitch outer loop detects it and the auto-thrust system changes the sink rate
c. The pitch inner loop detects it and the pitch outer loop changes the sink rate

185. The glide path is being intercepted from below and the angular deviation is decreasing — how does the pitch controller react after synchronization? ●

a. With an increasing pitch-up
b. With an increasing pitch-over
c. With a constant pitch attitude until the beam centre

186. An aircraft intercepts the glide path from above — what does the pitch controller command at synchronization? ●

a. A pitch-up, so the aircraft levels off above the glide path
b. A pitch-over, so the aircraft flares onto the glide path
c. A pitch-up, so the aircraft flares onto the glide path

Answers — Approach and land mode, glideslope submode

154. a The take off mode is listed as a flight director function only. (L1, PDF p. 200)

155. a Four hundred feet above ground is the height given for this lock-out. (L1, PDF p. 200)

156. b Five hundred feet is the height given with the JAR-AWO reference for this restriction. (L1, PDF p. 202)

157. b The armed condition is annunciated as G/P - ARM. (L1, PDF p. 204)

158. a The printed usable range of the glide path signal is plus or minus 0,5 degrees. (L1, PDF p. 204)

159. b Below fifteen hundred feet AGL the angular deviation can no longer be used directly, because of beam convergence. (L1, PDF p. 206)

160. b The vertical beam sensor watches the glide path deviation from the moment the mode is selected. (L1, PDF p. 208)

161. c Capture is triggered when the angular deviation falls below 0,2 degrees. (L1, PDF p. 208)

162. b Below 400 ft the button press no longer deactivates the mode. (L2, PDF p. 200)

163. b A precision approach is defined by a glide path indicator that permanently shows deviation from the intended glide path. (L2, PDF p. 200)

164. c The absence of vertical guidance is what makes an approach non-precision. (L2, PDF p. 200)

165. a Charted missed approach procedures or ATC guidance are the two printed sources. (L2, PDF p. 200)

166. b The two names stand for one and the same mode. (L2, PDF p. 200)

167. c The mode runs from localizer intercept through glide path intercept to roll out. (L2, PDF p. 202)

168. b Only a Go Around selection or switching the autopilot off removes the mode below that height. (L2, PDF p. 202)

169. c Both the ILS frequency and the runway heading at the PSC knob have to be set first. (L2, PDF p. 204)

170. a At about -0,2° the altitude hold drops out and the aircraft begins descending onto the beam. (L2, PDF p. 204)

171. c An angular deviation of about -0,2° corresponds roughly to half a dot below on the glide path deviation indicator. (L2, PDF p. 204)

172. c Ground reflection spoils the glide path signal, so control is dropped by about 70 ft. (L2, PDF p. 204)

173. c The same vertical offset subtends a larger angle as the range closes. (L2, PDF p. 206)

174. a The effect of the permanently rising loop amplification is called beam convergence. (L2, PDF p. 206)

175. b The first method combines the angular deviation with radio height. (L2, PDF p. 206)

176. c Sink rate feedback closes the inner loop. (L2, PDF p. 208)

177. b Inherent stability is the printed reason for the steady sink rate. (L2, PDF p. 208)

178. b Go Around can supersede Approach/Land at any height, unlike other upper modes. (L3, PDF p. 200)

179. b The sink rate at that moment is frozen until the flare starts. (L3, PDF p. 204)

180. c A category II only autopilot comes out at 100 ft and the pilot flares and lands. (L3, PDF p. 204)

181. c Low gain far out makes it sluggish, and rising gain close in makes it oscillate. (L3, PDF p. 206)

182. c Without a true slant range the deviation cannot be computed exactly, so it is estimated. (L3, PDF p. 206)

183. c Passing the marker starts a time based estimate that relies on the known OM to transmitter distance. (L3, PDF p. 206)

184. a The outer loop sees the beam error and the inner loop carries out the sink rate change. (L3, PDF p. 208)

185. b Synchronization reverses what the raw sign would command, so the aircraft pitches over onto the beam. (L3, PDF p. 208)

186. c All signs invert from above, giving a pitch-up and a flare onto the beam. (L3, PDF p. 208)

Flare, pitch rollout and yaw align

PDF pages 210–221 · 47 questions

LEVEL 1 — RECALL

187. Below which height AGL does the auto-throttle or auto-thrust system reduce the thrust to idle ●

a. 15 ft
b. 50 ft
c. 120 ft

188. From which vertical speed to which vertical speed does the flare reduce the descent shortly before touchdown ●

a. From approx. -600 to -750 ft/min down to approx. -120 ft/min
b. From approx. -400 to -600 ft/min down to approx. -120 ft/min
c. From approx. -600 to -750 ft/min down to approx. -250 ft/min

189. The landing controller is compensating for the ground effect — how much is added to the actual radio altitude? ●

a. Approximately 15 ft
b. Approximately 20 ft
c. Approximately 50 ft

190. Which movement does the pitch rollout submode command directly after touch down on the main gear ●

a. A controlled decrease of the bank angle
b. A controlled increase of the pitch attitude
c. A controlled decrease of the pitch attitude

191. Which annunciation does the flight mode annunciator display once the nose gear is being lowered onto the runway ●

a. LAND
b. FLARE-ACTIVE
c. ROLLOUT

192. Which name is given to the angle between the aircraft heading and its track on a crosswind approach ●

a. Drift angle or lateral slip angle
b. Crab angle or wind correction angle
c. Course angle or bearing angle

193. An automatic landing is flown in a crosswind — below which radio height is the alignment carried out? ●

a. Below 15 ft
b. Below 30 ft
c. Below 50 ft

194. From where does the preselected course reach the yaw channel as its reference value ●

a. From the flight management computer
b. From the inertial reference system
c. From the autoflight control panel

LEVEL 2 — UNDERSTAND AND EXPLAIN

195. A wind correction angle has been carried down the approach — what does yaw attitude or heading control do during the auto land? ●

a. It aligns the ground track with the runway centre line
b. It aligns the longitudinal axis with the runway direction
c. It holds the wind correction angle until the nose gear is down

196. Which signal guides the aircraft along the runway centre line after touch-down ●

a. The glide path deviation signal
b. The radio altitude signal
c. The localizer deviation signal

197. Why does a zero sink rate have to be avoided during the flare ●

a. To assure a positive touchdown
b. To avoid a tail strike
c. To shorten the touch down zone

198. The flare window is being computed during an automatic landing — which two momentary quantities determine it? ●

a. Vertical speed and pitch attitude
b. Radio height and ground speed
c. Vertical speed and ground speed

199. During the flare, which pair of quantities does the controller use to calculate the path to be flown down to the touch down

a. The momentary radio altitude and the derived crab angle
b. The momentary pitch attitude and the derived ground speed
c. The momentary landing gear height and the derived sink rate

200. Which change to the calculated landing path would the ground effect produce if it were left uncompensated ●

a. A steepening of the path
b. A shortening of the path
c. A flattening out of the path

201. Which change behind the wing produces the ground effect as the aircraft approaches the ground ●

a. The angle of sweep decreases
b. The angle of down wash increases
c. The angle of down wash decreases

202. Which moment follows from the increased arm to the centre of gravity together with the increased lift at the horizontal stabilizer ●

a. A pitch up moment
b. A pitch down moment
c. A yawing moment

203. Which two modes must already be active as the precondition for a flare activation ●

a. Glideslope as the vertical and localizer as the lateral mode
b. Altitude hold as the vertical and heading select as the lateral mode
c. Localizer as the vertical and glideslope as the lateral mode

204. The main gear has just touched down at the end of an automatic landing — which capability is lost at that moment? ●

a. An automatic go around manoeuvre
b. An automatic rudder alignment
c. An automatic deceleration process

205. Which part of the autoflight system switches itself off automatically at the touch down of the main gear ●

a. The autopilot roll channel
b. The auto-throttle
c. The autopilot yaw channel

206. Which signal is produced when the actual vertical speed is subtracted from the commanded vertical speed

a. The error signal for the pitch inner loop
b. The reference signal for the pitch outer loop
c. The feedback signal for the trim integrator

207. The flare detector starts the flare 50 ft above the ground at the earliest. Which quantity modifies that value? ●

a. The derived vertical speed
b. The H1 height bias
c. The ground speed

208. The flare detector rounds the flight path off continuously — which limit does it observe? ●

a. The upper pitch attitude limit
b. The upper sink rate limit
c. The upper bank angle limit

209. The remaining lift has to be killed after the touch down — which two actions do that?

a. Derotation and ground spoiler extension
b. Auto brake application and ground spoiler extension
c. Derotation and reverse thrust selection

210. Why does the aircraft heading have to be brought onto the runway heading shortly before touch down ●

a. The landing gear is aligned with the longitudinal axis
b. The nose wheel steering is available right after touch down
c. The rudder loses its effectiveness right after touch down

211. The decrab is being flown on an automatic landing — which reference is the heading aligned with? ●

a. The localizer deviation
b. The present track
c. The preselected course

212. The preselected course is the reference for the align phase — what must the pilot check before the alignment? ●

a. That it agrees with the localizer frequency
b. That it agrees with the present heading
c. That it agrees with the runway heading

213. Which channel and which surface does the align phase require ●

a. A yaw channel acting on the rudder
b. A yaw channel acting on the nose wheel
c. A roll channel acting on the ailerons

214. Which axis receives a coordinated input during the alignment, depending on the type of aircraft

a. The pitch axis
b. The vertical axis
c. The roll axis

215. The flight director is on when the yaw align phase begins — what changes on the display? ●

a. The pitch bar is replaced by a yaw bar
b. The roll bar is simply removed from view
c. The roll bar is replaced by a yaw bar

216. A crosswind automatic landing is flown — what does the flight mode annunciator show as the yaw align phase begins? ●

a. Nothing at all is annunciated for it
b. ROLLOUT appears as the lateral mode
c. ALIGN appears as the lateral mode

217. The align controller needs the present heading — where does it come from?

a. A magnetic heading system or a radio altimeter
b. A localizer receiver or an inertial reference system
c. A magnetic heading system or an inertial reference system

LEVEL 3 — APPLY AND REASON

218. An aircraft is landing with a tailwind, which has increased its sink rate — what does this do to the flare window?

a. It leaves the flare height unchanged but steepens the flare
b. It decreases the height at which the flare begins
c. It increases the height at which the flare begins

219. What does the simplicity of the flare controller have to be traded off against

a. A relatively narrow touch down zone and a large variation in sink rate
b. A relatively wide touch down zone and a large variation in sink rate
c. A relatively wide touch down zone and a large variation in pitch attitude

220. An aircraft is set down so softly that the weight on wheels signal is late — what does that cost on the landing roll?

a. The ground spoilers extend early
b. The braking action is delayed
c. The reverse thrust is inhibited

221. Which property of the tyres impedes the directional control by the landing gear after a greaser landing

a. They can accept only a limited rolling velocity
b. They can accept only a limited lateral force
c. They can accept only a limited vertical load

222. The landing controller adds a height bias to the measured radio altitude close to the ground — what does this achieve at the wheels? ●

a. A pitch up is still commanded once the aircraft is on the ground
b. A sink rate is still commanded once the aircraft is on the ground
c. A thrust increase is still commanded once the aircraft is on the ground

223. An aircraft enters ground effect in the last feet of the flare — which set of changes follows? ●

a. More lift, less induced drag and the centre of lift further aft
b. Less lift, less induced drag and the centre of lift further aft
c. More lift, more induced drag and the centre of lift further forward

224. A heavy aircraft carries a second set of antennas at the nose landing gear — what is switched over once the gear is extended? ●

a. The localizer receivers go to the nose gear antennas
b. The radio altimeters go to the nose gear antennas
c. The glide path receivers go to the nose gear antennas

225. Which result is achieved by feeding the glide path receivers from the nose gear antennas on a heavy aircraft ●

a. It gets the same flare window as a smaller aircraft
b. It gets the same approach speed as a smaller aircraft
c. It gets a lower flare window than a smaller aircraft

226. The flare controller stops rounding the path off at its upper limit — what would happen beyond that limit? ●

a. The aircraft would risk a long float before touch down
b. The aircraft would risk a tail strike at touch down
c. The aircraft would risk a nose gear strike at touch down

227. Why does a vertical acceleration signal damp the activity of the flare controller

a. It lets the radio altitude be filtered smoothly
b. It lets disturbances be recognised early
c. It lets the sink rate be measured accurately

228. Why does the remaining component of lift have to be decreased after the touch down

a. To increase rudder authority and yaw damping
b. To increase runway adhesion and brake efficiency
c. To increase tyre life and brake cooling

229. The auto-trim is switched off at the same time as the derotation — what would otherwise drive the trim? ●

a. A constant pitch error caused by the stabilizer's positive moment
b. A constant pitch error caused by the elevator's negative moment
c. A constant pitch rate caused by the elevator's negative moment

230. Why is the aircraft headed into the wind during a cross wind approach ●

a. To prevent a bank angle on the localizer beam
b. To prevent a displacement from the localizer beam
c. To prevent a lateral load on the landing gear

231. Why is the alignment left until shortly before the touch down ●

- **a.** It minimizes the lateral drift
- **b.** It minimizes the rudder deflection
- **c.** It minimizes the crosswind component

232. Which comparison produces the control error used by the align controller

- **a.** The present heading against the present track
- **b.** The localizer deviation against the runway heading
- **c.** The present heading against the runway heading

233. An aircraft on an automatic approach is in landing configuration and passing 200 ft radio height — what is the state of the align phase?

- **a.** Neither armed nor active
- **b.** Armed but not yet active
- **c.** Active and holding the runway heading

Answers — Flare, pitch rollout and yaw align

187. a The printed height for thrust reduction to idle is 15 ft AGL. (L1, PDF p. 210)

188. a Those are the printed glide path and touchdown sink rates. (L1, PDF p. 210)

189. b About 20 ft of bias is added to the measured radio height. (L1, PDF p. 212)

190. c The submode lowers the nose by reducing the pitch attitude in a controlled way. (L1, PDF p. 216)

191. c ROLLOUT is shown once the pitch rollout submode runs. (L1, PDF p. 216)

192. b That angle is the crab angle, also called the wind correction angle. (L1, PDF p. 218)

193. b Alignment is carried out below 30 ft of radio height. (L1, PDF p. 218)

194. c The crew sets the preselected course on the autoflight control panel and that is the yaw reference. (L1, PDF p. 220)

195. b The crab is taken out by yawing the aircraft onto the runway direction. (L2, PDF p. 210)

196. c Roll out steering uses the localizer deviation. (L2, PDF p. 210)

197. a Some sink rate must remain so that the aircraft actually touches down. (L2, PDF p. 210)

198. c The flare window is computed from the momentary vertical speed and ground speed. (L2, PDF p. 210)

199. c The landing path is built from the gear height over the runway together with the derived sink rate. (L2, PDF p. 212)

200. c Extra lift in ground effect reduces the sink rate, so the path would flatten out. (L2, PDF p. 212)

201. c Near the ground the down wash angle falls off, which raises the effective angle of incidence. (L2, PDF p. 212)

202. b More tail lift on a longer arm pushes the nose down. (L2, PDF p. 212)

203. a The flare only arms out of a coupled approach with the glideslope active vertically and the localizer active laterally. (L2, PDF p. 212)

204. a Once the main gear is down the automatic go around can no longer be flown. (L2, PDF p. 212)

205. b The autothrottle switches itself off at touchdown while the yaw channel carries on through the roll out. (L2, PDF p. 212)

206. a That subtraction is what generates the error fed to the pitch inner loop. (L2, PDF p. 214)

207. b The earliest flare point is the 50 ft main gear height modified by the H1 height bias that forces a positive touch down. (L2, PDF p. 214)

208. a The rounding off is bounded by the upper pitch attitude limit. (L2, PDF p. 214)

209. a Lowering the nose and putting the ground spoilers out are the two actions named. (L2, PDF p. 216)

210. a The wheels point where the fuselage points, so any crab left on would take the aircraft off the runway. (L2, PDF p. 218)

211. c The align phase flies the heading onto the preselected course. (L2, PDF p. 218)

212. c The align phase is only as good as the preselected course, so it must match the runway heading. (L2, PDF p. 218)

213. a The decrab is flown by the yaw channel through the rudder. (L2, PDF p. 218)

214. c Some aircraft add a coordinated roll input, the low wing method, with the decrab. (L2, PDF p. 218)

215. c The roll bar gives way to a yaw bar as the align phase starts. (L2, PDF p. 218)

216. a The align phase gets no annunciation at all. (L2, PDF p. 218)

217. c Present heading comes either from the compass system or from the inertial system. (L2, PDF p. 220)

218. c A faster descent raises the flare height so that there is more time to arrest the sink rate. (L3, PDF p. 210)

219. b The price of the simple law is a wide touchdown zone and a spread of touchdown sink rates. (L3, PDF p. 210)

220. b A late weight on wheels signal or a late wheel spin up both hold up the braking. (L3, PDF p. 212)

221. b The tyres can take only a limited lateral force, so the landing gear cannot steer the aircraft. (L3, PDF p. 212)

222. b Because the controller thinks it is higher than it is, it keeps commanding a descent and so forces a positive touch down. (L3, PDF p. 212)

223. a Ground effect gives more lift, less induced drag and moves the centre of lift rearwards. (L3, PDF p. 212)

224. c Gear extension switches the glide path receivers over to the antennas on the nose gear. (L3, PDF p. 212)

225. a Measuring from the nose gear lifts the aircraft virtually on the beam so the flare window matches a small aircraft's. (L3, PDF p. 212)

226. b Too much nose up at touchdown would put the tail on the runway. (L3, PDF p. 214)

227. b Seeing a disturbance early means the controller does not have to react violently later. (L3, PDF p. 214)

228. b Killing the lift puts weight on the wheels so they grip and the brakes work. (L3, PDF p. 216)

229. b The elevator holds a steady nose down moment after derotation, and the standing pitch error that leaves would wake the trim system up. (L3, PDF p. 216)

230. b Crabbing into the wind stops the aircraft being blown off the beam, so it drifts along it instead. (L3, PDF p. 218)

231. a Once the crab is taken out the aircraft starts drifting sideways, so the manoeuvre is kept short. (L3, PDF p. 218)

232. c The align loop subtracts the measured heading from the runway heading set as preselected course. (L3, PDF p. 220)

233. b The landing configuration arms the phase, but it only goes active below 30 ft, so at 200 ft it is still armed. (L3, PDF p. 221)

Yaw rollout submode

PDF pages 222–225 · 7 questions

LEVEL 1 — RECALL

234. An automatic landing is being flown onto the runway — which event starts the roll-out phase? ●

a. The compression of the nose gear
b. The extension of the ground spoilers
c. The touch down of the main gear

235. The aircraft is decelerating on the runway under the yaw channel — which event ends the roll-out phase? ●

a. The lowering of the nose gear
b. The disengagement of the autopilot
c. The aircraft coming to a stop

LEVEL 2 — UNDERSTAND AND EXPLAIN

236. Which means takes over the directional control increasingly after the derotation ●

a. The nose wheel steering
b. The rudder deflection
c. The differential braking

LEVEL 3 — APPLY AND REASON

237. Why can the localizer deviation signal be used for the yaw control during the roll-out ●

a. The localizer is aligned with the aircraft track
b. The localizer is aligned with the runway centre line
c. The localizer is aligned with the preselected course

238. The aircraft is rolling out fast with the nose gear still off the ground — how is the direction held? ●

a. By rudder deflections only
b. By rudder deflections and nose wheel steering
c. By nose wheel steering only

239. The aircraft is being steered along the centre line after landing — which two signals does the yaw controller process? ●

a. The crab angle and the localizer deviation
b. The heading error and the localizer deviation
c. The heading error and the glide path deviation

240. All three gears are on the runway during the roll-out — what else does the yaw channel now control?

a. The rudder trim actuator
b. The main wheel brakes
c. The nose wheel steering

Answers — Yaw rollout submode

234. c Roll out begins the moment the main gear is on the runway. (L1, PDF p. 222)

235. b Roll out runs until the crew disengages the autopilot. (L1, PDF p. 222)

236. a Once the nose wheel is down it progressively takes over the steering. (L2, PDF p. 222)

237. b Because the beam lies along the centre line, a beam deviation is a centre line deviation. (L3, PDF p. 222)

238. a With the nose off the ground only the rudder is available, and at that speed it still bites. (L3, PDF p. 222)

239. b Both a heading error and a lateral displacement have to be removed, so both signals are used. (L3, PDF p. 224)

240. c Once the nose gear is down the same yaw channel steers it as well as the rudder. (L3, PDF p. 225)

Go-around mode

PDF pages 226–231 · 32 questions

LEVEL 1 — RECALL

241. Which name is given to the transition phase from an approach to a stabilized climb ●

a. A roll out manoeuvre
b. A go around manoeuvre
c. An align manoeuvre

242. Which channel takes over the vertical guidance once the go around manoeuvre has started ●

a. The yaw channel
b. The roll channel
c. The pitch channel

243. Which form of control does the first go around concept activate at the start of the manoeuvre ●

a. Vertical speed control to a fixed climb value
b. Pitch attitude control to a fixed positive value
c. Airspeed control to a fixed reference value

LEVEL 2 — UNDERSTAND AND EXPLAIN

244. A go around is selected on the approach — which three parts of the autoflight system come in together? ●

a. The roll channel, the yaw channel and the auto-thrust system
b. The pitch channel, the roll channel and the auto-thrust system
c. The pitch channel, the yaw channel and the auto-thrust system

245. The crew changes the slat and flap setting during a go around — what happens to the airspeed reference values?

a. They are adapted automatically
b. They are frozen at the previous value
c. They have to be reselected by hand

246. A go around is being flown — what is the minimum climb rate that has to be observed? ●

a. 600 ft/min
b. 120 ft/min
c. 1200 ft/min

247. Which minimum flight path angle has to be observed during a go around ●

a. + 0.5°
b. + 18°
c. + 1.5°

248. Why does the airworthiness standard demand a minimum climb rate and a minimum flight path angle ●

a. To achieve obstacle clearance
b. To achieve a stabilized airspeed
c. To achieve a positive rate of climb

249. Which condition must be satisfied before the go around mode may be terminated ●

a. A minimum airspeed above the reference has to be passed
b. A minimum height above the ground has to be passed
c. A minimum time in thrust mode has to elapse

250. Which thrust does the auto-throttle command when the go around mode becomes active ●

a. The maximum continuous thrust available in the climb
b. The maximum thrust available for five minutes
c. The reduced climb thrust available after flap retraction

251. Which track does the roll channel maintain during the go around ●

a. The track of the runway centre line extended
b. The track selected on the autoflight control panel
c. The track it was following at the start of the manoeuvre

252. Which penalty follows the inversion of a sink rate into a climb rate during the go around ●

a. The thrust is reduced
b. The airspeed is reduced
c. The pitch attitude is reduced

253. Between which climb rates does the transition from attitude guidance to speed guidance take place ●

a. 1200 ft/min and 1800 ft/min
b. 120 ft/min and 600 ft/min
c. 600 ft/min and 1200 ft/min

254. Which flight condition makes the climb rate drop drastically during a go around ●

a. A thrust reduction or a bank
b. A flap or a gear retraction
c. Wind shear or a downdraft

255. Which principle is the speed reference control law built on ●

a. Total pressure management
b. Total thrust management
c. Total energy management

256. Which two modes is the speed reference control law used for

a. The flight director take-off mode and the go around mode
b. The flight director land mode and the go around mode
c. The flight director take-off mode and the flare mode

257. Which quantity is the potential flight path angle directly proportional to ●

a. The pitch attitude on the actual flight path
b. The acceleration on the actual flight path
c. The thrust setting on the actual flight path

258. Which property of the aircraft does the potential flight path angle reflect

a. Its total energy
b. Its kinetic energy only
c. Its potential energy only

259. Which quantity is the correction angle proportional to ●

a. The speed error
b. The angle of attack
c. The flight path error

260. Which signal is formed at the last summing point and used as the reference for the pitch attitude controller ●

a. The sum of the flight path error and the correction angle
b. The difference of the actual and potential flight path angles
c. The sum of the pitch attitude and the angle of attack

LEVEL 3 — APPLY AND REASON

261. A large thrust increase at the start of a go around produces a strong nose up moment — which limits must the go around controller watch? ●

a. The flap or landing gear speed limits
b. The pitch attitude or angle of attack limits
c. The bank angle or sideslip limits

262. An aircraft has an auto-thrust system and the crew wants a go around — what do they do with the power levers? ●

a. Pull them back to the idle detent
b. Push them to the TOGA detent
c. Press the TOGA switch on them

263. Why is the go around trigger tied to uncompressed gears with slats and flaps extended

a. It prevents triggering during other flight phases
b. It prevents triggering below the minimum height
c. It prevents triggering with the autopilot engaged

264. The aircraft touches down while a go around is being flown — what happens to the mode? ●

a. It changes into the rollout mode
b. It stays active
c. It is terminated at once

265. The crew decides to level off and fly a radar heading after a go around — how is the mode taken out? ●

a. By selecting other upper modes for pitch and roll together
b. By disengaging the autopilot and flying by hand
c. By selecting other upper modes for pitch and roll independently

266. The go around controller rotates the aircraft to a fixed positive pitch attitude — what follows from that? ●

a. Lift increases and the momentary sink rate decreases
b. Lift increases and the momentary airspeed increases
c. Drag increases and the momentary sink rate decreases

267. Which control replaces pitch attitude hold a short while after the start of the go around ●

a. Thrust control of the auto-throttle system
b. Airspeed control of the pitch channel
c. Vertical speed control of the pitch channel

268. Airspeed control is running in a go around and the climb rate falls below 1200 ft/min — what does the vertical guidance do? ●

a. It reverts to vertical speed control
b. It stays on airspeed control
c. It reverts to pitch attitude control

269. Why does the aircraft keep climbing once the pitch controller holds a constant airspeed

a. The surplus of go around thrust gives a positive flight path angle
b. The surplus of go around thrust gives a positive pitch attitude
c. The surplus of lift gives a positive flight path angle

270. An aircraft is flying on a path whose angle equals its potential flight path angle — what are its errors? ●

a. Airspeed error zero and flight path error zero
b. Airspeed error at a maximum and flight path error at a maximum
c. Airspeed error zero and flight path error not zero

271. Which angle appears when the airspeed reference is changed while the thrust is held constant ●

a. An error angle that changes the present angle of attack
b. A demand angle that changes the present pitch attitude
c. A correction angle that changes the present flight path angle

272. Go around is selected on an approach — which speed is captured in the memory as the speed target?

a. The actual airspeed at that moment
b. The minimum selectable speed
c. The approach reference speed

Answers — Go-around mode

241. b That transition is exactly what is called a go around. (L1, PDF p. 226)

242. c Vertical guidance in a go around belongs to the pitch channel. (L1, PDF p. 226)

243. b The first concept rotates to and then holds a fixed nose up attitude. (L1, PDF p. 228)

244. b Go around is a common mode and brings pitch, roll and the thrust system in together. (L2, PDF p. 226)

245. a The references follow the configuration without any crew action. (L2, PDF p. 226)

246. b The minimum climb rate for the go around is 120 ft/min. (L2, PDF p. 226)

247. a Half a degree is the minimum flight path angle. (L2, PDF p. 226)

248. a The minima exist so the aircraft will clear what is ahead of it. (L2, PDF p. 226)

249. b The mode is locked in until a minimum height above the ground has been exceeded. (L2, PDF p. 226)

250. b Go around thrust is the five minute maximum rating. (L2, PDF p. 226)

251. c The roll channel simply holds whatever track was being flown when the go around began. (L2, PDF p. 226)

252. b Turning a descent into a climb is paid for in speed. (L2, PDF p. 228)

253. c The changeover is spread continuously over 600 to 1200 ft/min of climb. (L2, PDF p. 228)

254. c Wind shear or a downdraft is the case given for a sudden loss of climb rate. (L2, PDF p. 228)

255. c The alternative concept works from the total energy of the aircraft. (L2, PDF p. 228)

256. a The same law serves the take-off director and the go around. (L2, PDF p. 228)

257. b It is the acceleration along the path that the potential angle follows. (L2, PDF p. 230)

258. a The potential angle is a measure of the whole energy state, not one part of it. (L2, PDF p. 230)

259. a The correction angle is sized by the speed error. (L2, PDF p. 230)

260. a S3 adds the correction angle onto the flight path error and the sum is the demand given to the pitch controller. (L2, PDF p. 230)

261. b The required thrust increase causes a large positive pitch moment, so the pitch attitude and angle of attack limits have to be observed. (L3, PDF p. 226)

262. b On an auto-thrust aircraft the levers are pushed into the TOGA detent. (L3, PDF p. 226)

263. a The configuration test makes sure the mode can only come up where it belongs, on an approach. (L3, PDF p. 226)

264. b The mode survives a touch down, which is what makes a touch and go possible. (L3, PDF p. 226)

265. c Picking a new upper mode ends the go around, and the vertical and lateral sides can be ended separately. (L3, PDF p. 226)

266. a More attitude means more lift, and more lift arrests the descent. (L3, PDF p. 228)

267. b Because the attitude hold is eating the speed, the pitch channel goes over to controlling airspeed. (L3, PDF p. 228)

268. c Losing climb performance sends the channel back to holding attitude so lift is restored. (L3, PDF p. 228)

269. a With the speed pegged, the spare thrust has nowhere to go but into climb angle. (L3, PDF p. 228)

270. a Flying at the potential angle means the energy is exactly balanced, so both errors vanish. (L3, PDF p. 230)

271. c The speed error is worked off by a correction angle that tilts the flight path so the aircraft accelerates or decelerates. (L3, PDF p. 230)

272. a Selecting go around closes the switch that freezes the actual airspeed into the memory as the target. (L3, PDF p. 231)